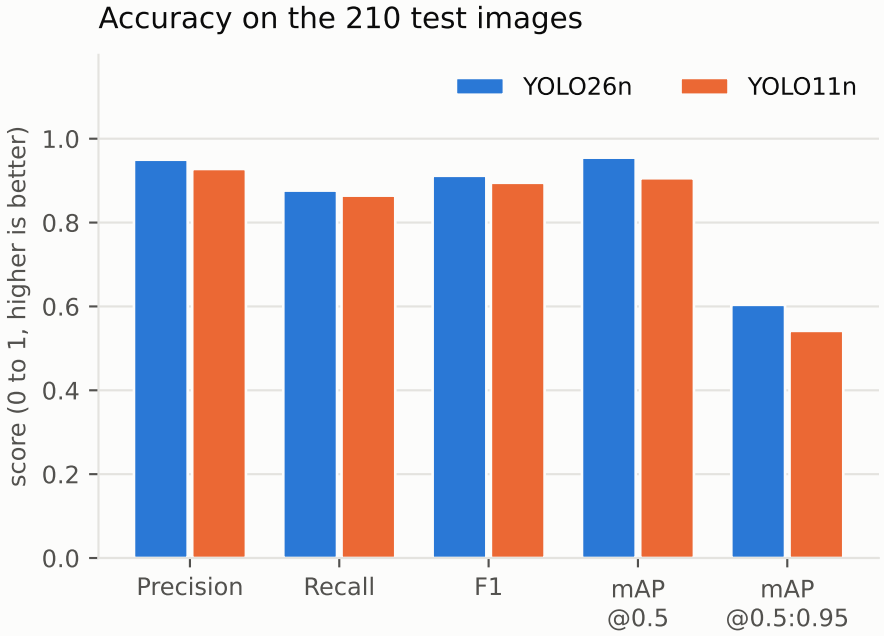
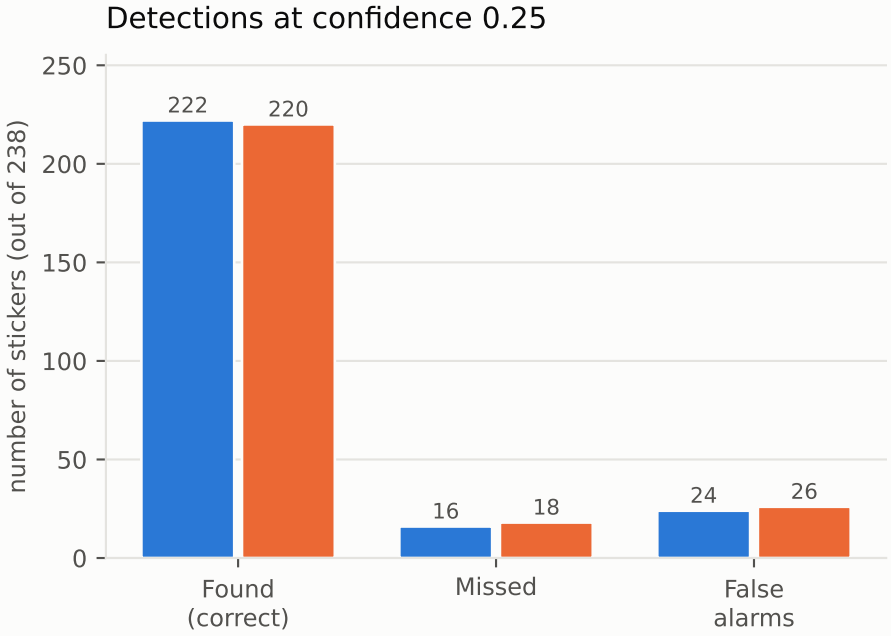
YOLO26n vs YOLO11n

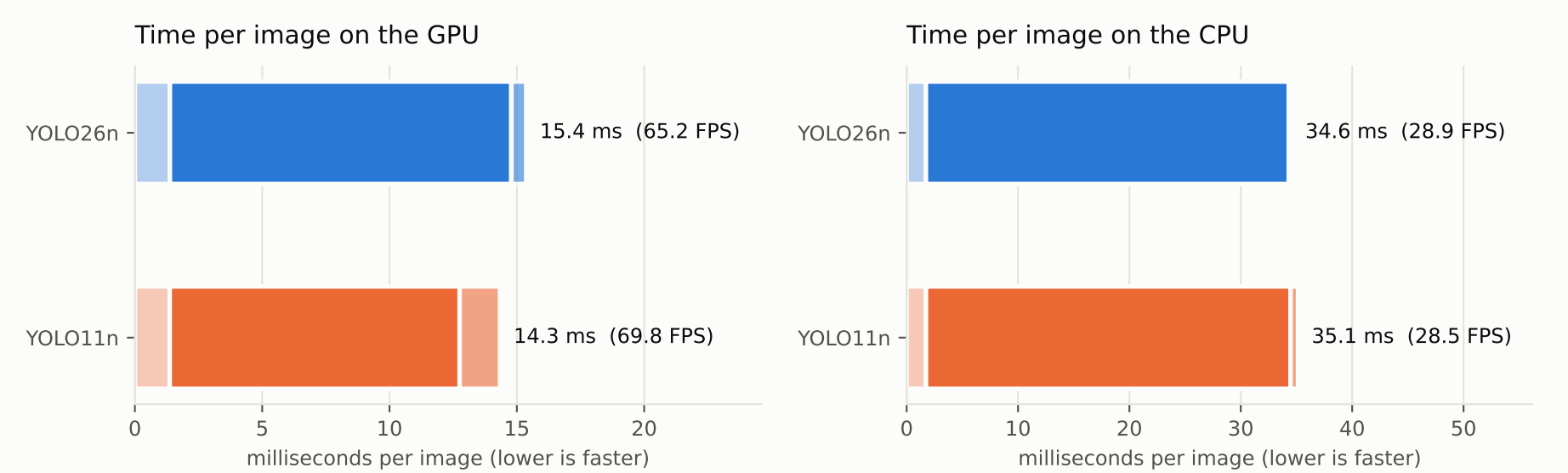
PSU Vehicle Sticker Detection Benchmark

November 10, 2026

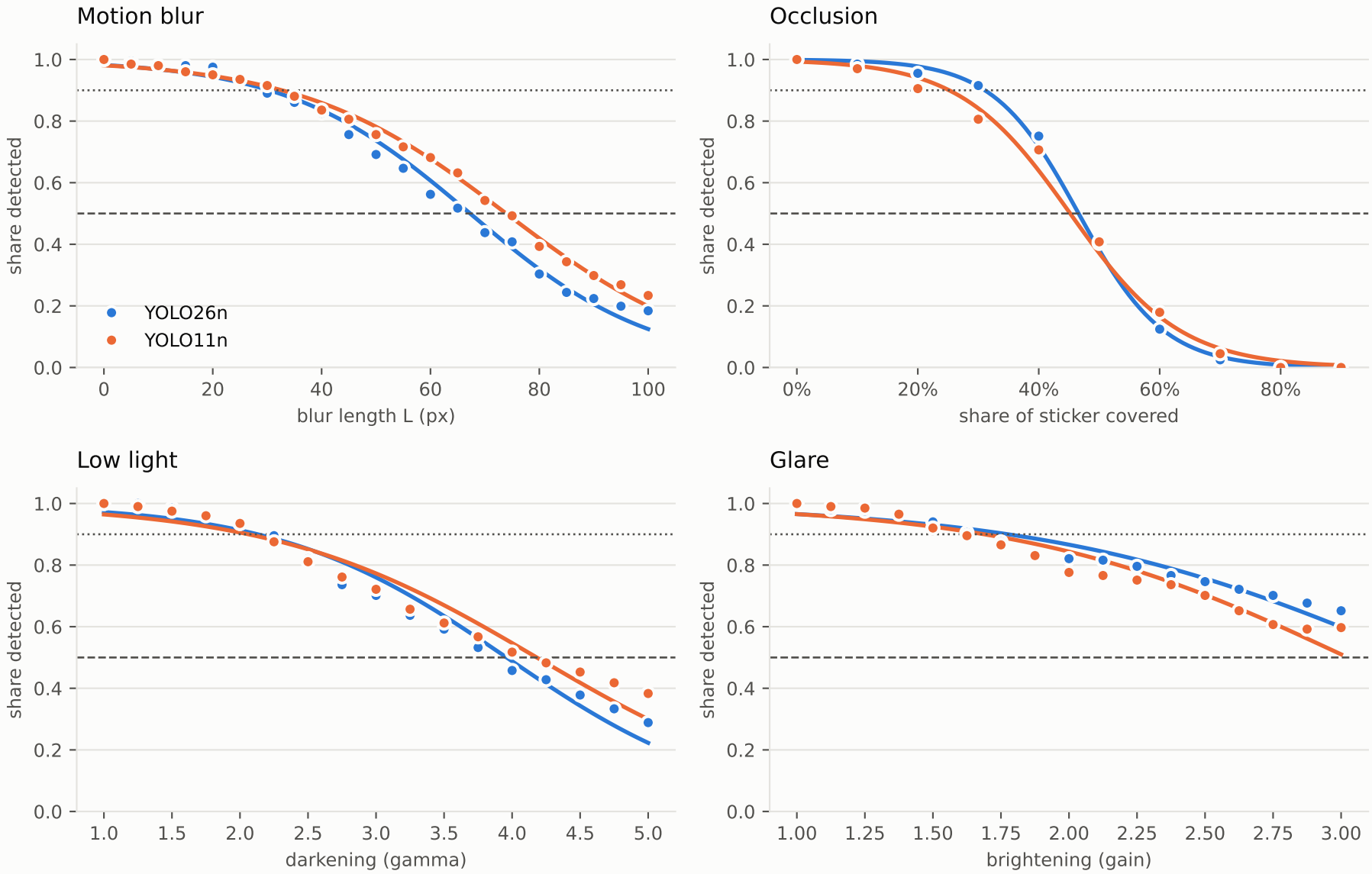
Measure	YOLO26n	YOLO11n	Better
Accuracy: mAP@0.5:0.95	0.604	0.542	YOLO26n
Accuracy: mAP@0.5	0.955	0.906	YOLO26n
Precision	0.950	0.928	YOLO26n
Recall	0.876	0.864	YOLO26n
Speed on GPU (FPS)	65.2	69.8	YOLO11n
Speed on CPU (FPS)	28.9	28.5	YOLO26n
Motion blur: breaks at (50%)	67 px	74 px	YOLO11n
Occlusion: breaks at (50%)	47% covered	45% covered	No clear difference (YOLO26n slightly
Low light: breaks at (50%)	gamma 3.96	gamma 4.18	YOLO11n
Glare: breaks at (50%)	gain 3.27x	gain 3.02x	YOLO26n


Accuracy on the 210 test images
YOLO26n
YOLO11n
score (0 to 1, higher is better)
1.0
0.8
0.6
0.4
0.2
0.0
Precision
Recall
F1
mAP
@0.5
mAP
@0.5:0.95

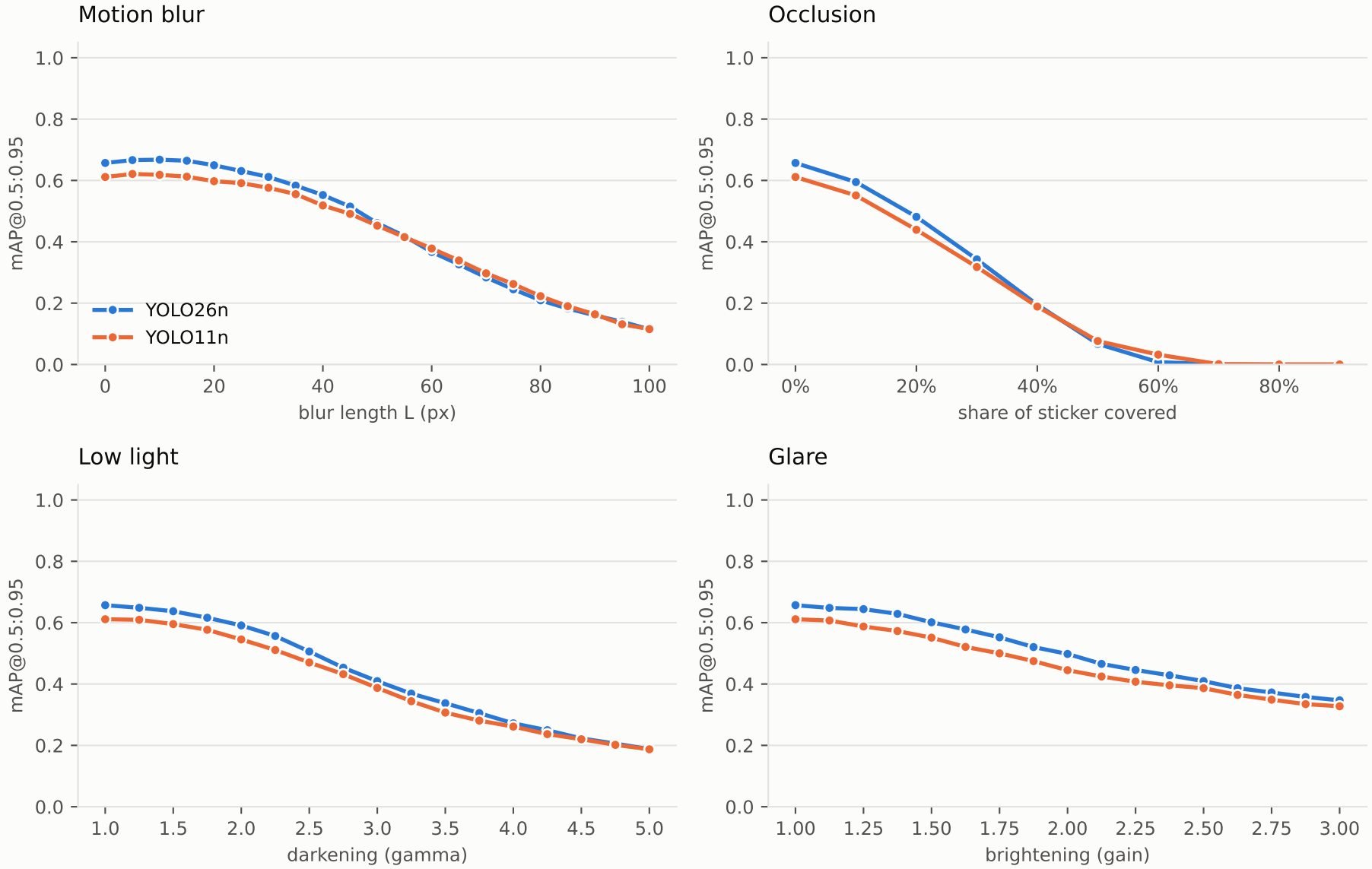

Detections at confidence 0.25
number of stickers (out of 238)
250
200
150
100
50
0
222
220
16
18
24
26
Found
(correct)
Missed
False
alarms



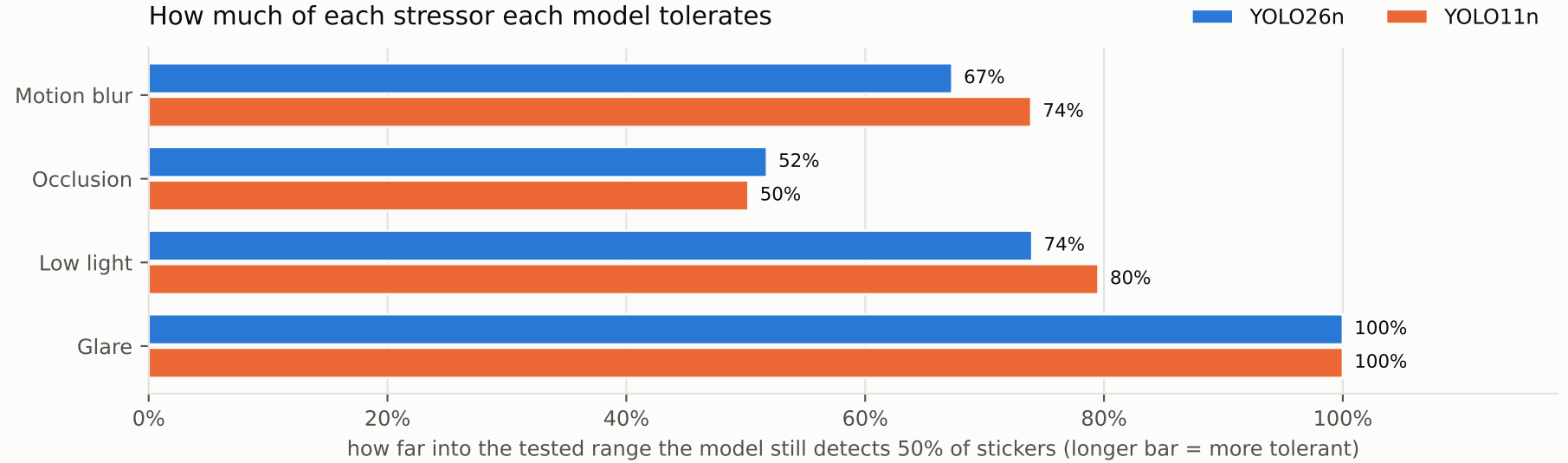
Bar shades, light to dark to medium: prepare image, model, cleanup (NMS). YOLO26n's cleanup is shorter because it is NMS-free.



Dots = measured, lines = fitted curve. Dashed line = breaking point (50% detected); dotted line = reliable detection (90%). Further right = harder.



Further right = harder.


How much of each stressor each model tolerates
YOLO26n
YOLO11n
Motion blur
67%
74%
Occlusion
52%
50%
Low light
74%
80%
Glare
100%
100%
0%
20%
40%
60%
80%
100%
how far into the tested range the model still detects 50% of stickers (longer bar = more tolerant)